

JAY BOMBATKAR

Embedded Systems Engineer | PCB Design | Firmware Development
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PROFESSIONAL SUMMARY

Electronics and Telecommunication Engineering final-year student (SGPA: 8.85) with hands-on experience in embedded systems design, PCB layout, and microcontroller firmware development. Proficient in STM32, PIC microcontrollers, communication protocols (UART, SPI, I2C, CAN Bus), and PCB design tools (KiCad, Altium Designer). Experienced in hardware validation, sensor testing, and debugging with oscilloscopes. Seeking an entry-level role in Embedded Systems Engineering, Hardware Design, or Electronics R&D.

TECHNICAL SKILLS

Programming Languages: Embedded C, Python, SQL

Microcontrollers & RTOS: STM32, PIC, FreeRTOS, Interrupts, Timer Programming, Memory Interfacing

Communication Protocols: UART, SPI, I2C, USB, CAN Bus, RS-485

PCB Design Tools: Altium Designer, KiCad (Schematic Capture, PCB Layout, Routing, DRC, Gerber Generation)

Debugging & Test Equipment: Oscilloscope, Multimeter

Simulation: LTspice (Circuit Simulation, Waveform Analysis)

Development Tools: MPLAB X IDE, MPLAB IPE, CCS Programmer, Git, GitHub, Linux

Hardware Skills: Datasheet Analysis, Component Selection, Hardware Validation, Circuit Troubleshooting

WORK EXPERIENCE

Hardware Test Intern | SuryaLogix Pvt. Ltd. *Mar 2026 – May 2026*

- Performed functional and performance testing of environmental and solar sensors (temperature, humidity, solar irradiation) to validate accuracy and compliance with industry specifications
- Conducted PCB-level debugging and fault isolation on sensor circuits using oscilloscope and multimeter, improving defect detection efficiency
- Verified analog and digital sensor outputs across varied operating conditions, ensuring robust field performance
- Executed production testing and quality assurance procedures in line with industry standards, contributing to zero-defect shipment targets
- Collaborated with cross-functional engineering team on root cause analysis, issue resolution, and continuous process improvement

PROJECTS

CNC Controller Using MPG (Manual Pulse Generator) Sponsored by Gaytri Kelkar CNC Pvt Ltd | Ongoing

- Developing a microcontroller-based CNC/EDM jogging controller using an MPG rotary encoder for precise manual axis control.
- Implemented single-axis configuration support for EDM machine operations with accurate positioning control.
- Designed and validated a 2-layer PCB in KiCad including schematic capture, routing, grounding planes, and EMI/EMC considerations.
- Developed Embedded C firmware for quadrature encoder interfacing, pulse generation, switch debouncing, and real-time motion control.
- Performed hardware testing and debugging using oscilloscope and MPLAB X IDE to ensure signal integrity and system reliability.

STM32-Based Embedded System PCB Design (Altium Designer)

- Designed a production-ready STM32 microcontroller PCB using Altium Designer, encompassing full schematic capture, Four-layer PCB layout, and design rule checks (DRC)
- Implemented regulated power supply circuitry, USB and UART communication interfaces, and peripheral interfacing for STM32 microcontroller

- Performed component placement optimization, differential pair routing, and impedance-controlled signal routing to ensure signal integrity and EMC compliance
- Generated complete fabrication output package including Gerber files, NC drill files, and Bill of Materials (BOM) for PCB manufacturing

Dual Output Power Supply PCB Design (KiCad)

- Designed and developed a dual-output regulated power supply PCB converting 12V DC input into stable 5V and 3.3V outputs using LD1117 regulators — directly applicable to embedded and power electronics applications.
- Implemented full schematic capture, PCB layout, routing, fuse protection, filtering capacitors, and LED indication circuits in KiCad.
- Generated Gerber files and BOM; validated the design for manufacturing readiness and compliance with power electronics standards.

- **EDUCATION**

Bachelor of Engineering – Electronics and Telecommunication

2022 – 2026

Savitribai Phule Pune University | SGPA: 8.85 / 10.0

CERTIFICATIONS

- PCB Design with KiCad – Udemy
- Hardware Design using PIC Microcontroller – Udemy

RELEVANT COURSEWORK

Analog Electronics | Digital Electronics | Microcontrollers & Embedded Systems | Communication Systems | Control Systems

ACHIEVEMENTS & LEADERSHIP

- Led team as Project Lead at Project Expo 2026 and 2025 PVPIT Bavdhan – managed project delivery and stakeholder presentation
- Led team as Event Lead at Quiz Competition 2024, PVPIT Bavdhan – coordinated logistics and participant engagement for technical event